

Table 3. Estimation of the segmented regression (time interval for citing publications: 2012; time interval for cited references: from 1650 to 2012)

Parameter	Estimate	SE	95% confidence interval	% growth rate	Doubling time [year]
a ₁	1739.2*	1.90	1735.5 – 1742.9		
a ₂	1942.2*	0.86	1940.5 - 1943.9		
a ₃	2010.9*	0.16	2010.6 - 2011.2		
b ₁	0.003*	0.000	0.003 - 0.003	0.27%	253.9
b ₂	0.022*	0.000	0.021 - 0.022	2.19%	31.9
b ₃	0.088*	0.001	0.086 - 0.090	9.20%	7.9
b ₄	-1.310*	0.290	-1.874 - -0.745	-73.01%	-

Notes. R²=.99

*p<.05

Table 4. Estimation of the segmented regression for the natural sciences (time interval for citing publications: from 1980 to 2012; time interval for cited references: from 1650 to 2012)

Parameter	Estimate	SE	95% confidence interval	% growth rate	Doubling time [year]
a ₁	1718.3*	2.39	1713.6 - 1723.0		
a ₂	1925.5*	1.77	1922.0 - 1929.0		
a ₃	1999.2*	0.75	1998.1 - 2001.1		
b ₁	0.003*	0.000	0.003 - 0.003	0.29%	231.1
b ₂	0.029*	0.000	0.028 - 0.030	2.94%	23.9
b ₃	0.081*	0.002	0.077 - 0.084	8.37%	8.7
b ₄	-0.189*	0.025	-0.238 - -0.139	-14.20%	-

Notes. R²=.99

*p<.05

Table 5. Estimation of the segmented regression for medical and health sciences (time interval for citing publications: from 1980 to 2012; time interval for cited references: from 1650 to 2012)

Parameter	Estimate	SE	95% confidence interval	% growth rate	Doubling time [year]
a ₁	1753.5*	2.23	1749.1 – 1757.8		
a ₂	1928.6*	1.60	1925.4 - 1931.7		
a ₃	2002.1*	0.63	2000.9 - 2003.4		
b ₁	0.003*	0.000	0.003 - 0.003	0.28%	250.2
b ₂	0.031*	0.000	0.030 - 0.032	3.10%	22.3
b ₃	0.089*	0.002	0.086 - 0.093	9.35%	7.8
b ₄	-0.297*	0.038	-0.37 - -0.222	-25.67%	-

Notes. R²=.99

*p<.05

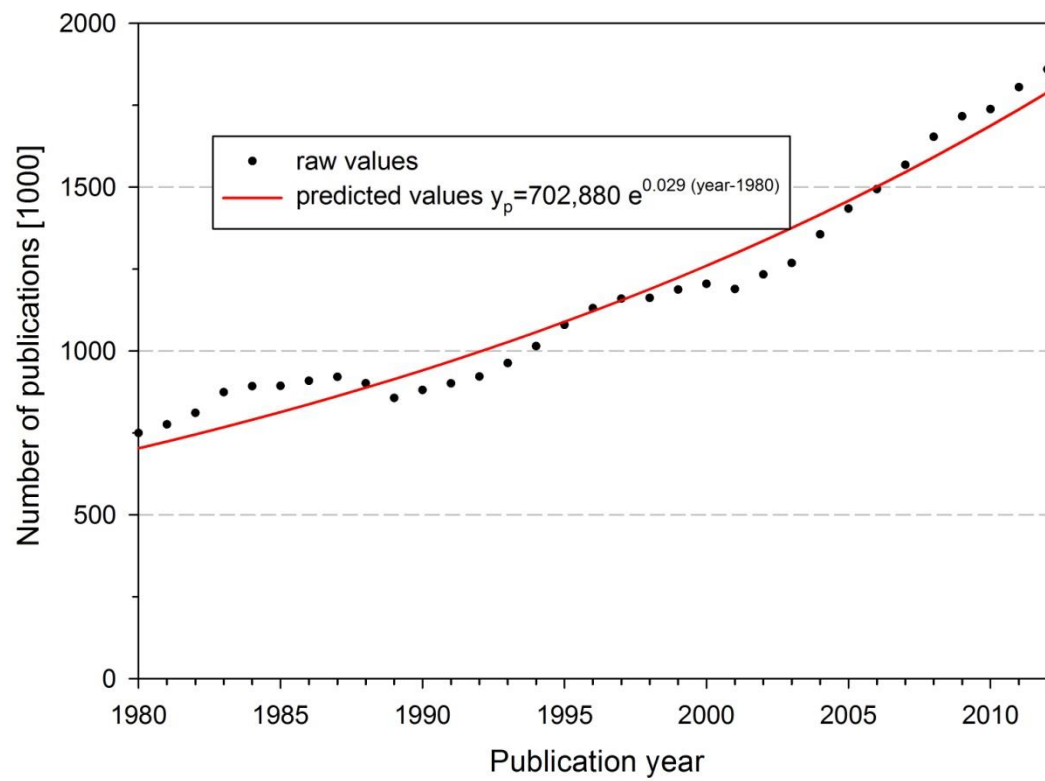


Figure 1. Exponential growth of scientific output from 1980 to 2012